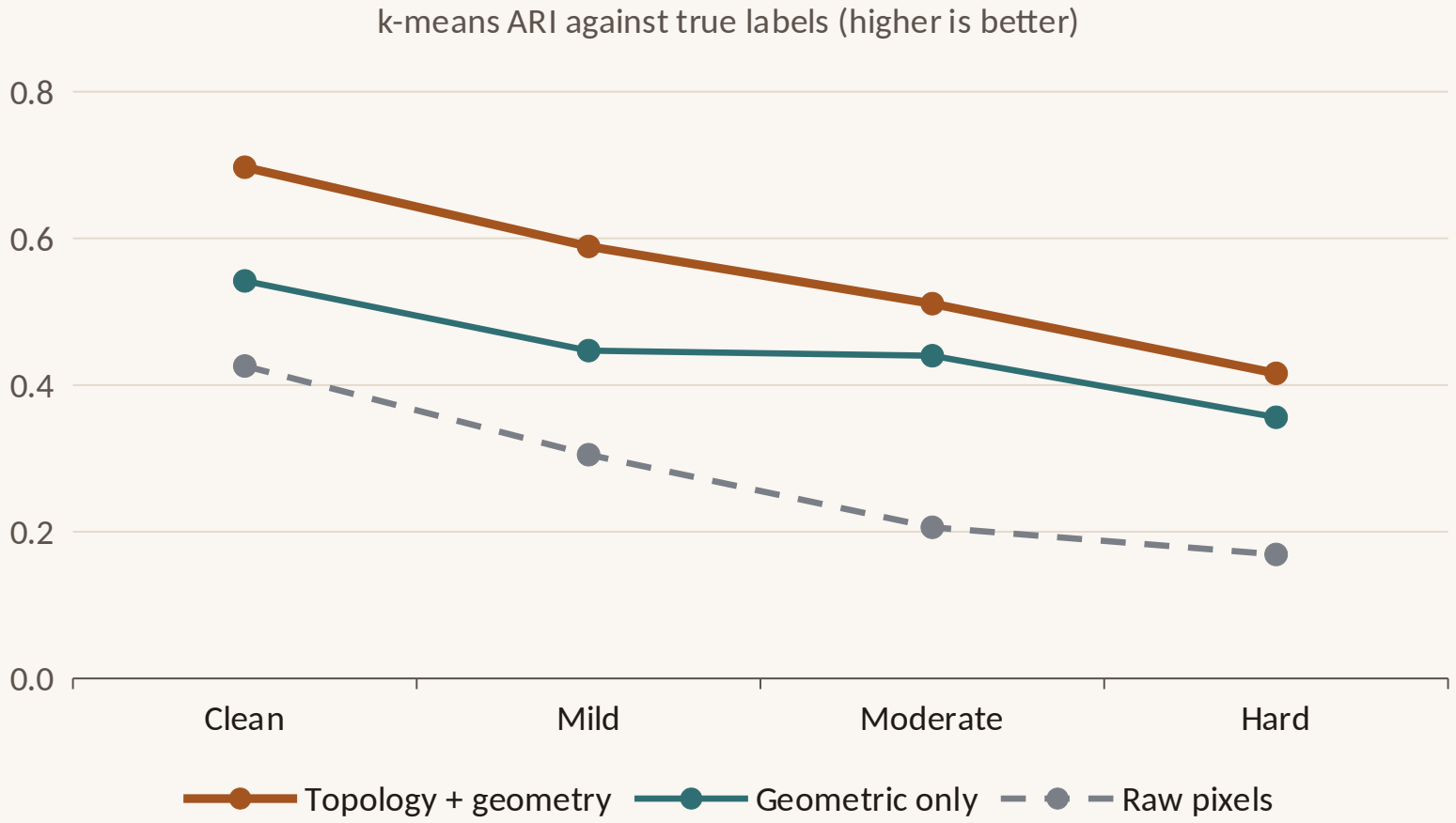
Clustering on raw pixels breaks as wafer variation grows



0.43 → 0.17

Raw pixels, clean to hard

0.54 → 0.36

Eight geometric features

0.70 → 0.42

Topology plus geometry (34 features)

Held-out seeds 2001–2003 · 540 balanced synthetic maps per dataset · k = 9 · mean of 3 seeds

Topology adds value alongside geometry, not on its own

Feature set (hard level)	k-means ARI	CV accuracy	Clean → hard accuracy
Raw pixels (PCA-20)	0.17	0.74	0.37
Geometric only (8)	0.36	0.73	0.32
Topology only (26)	0.36	0.71	0.39
Topology + geometry (34)	**0.42**	**0.80**	**0.43**

The threshold matters

At the default 0.1, topology plus geometry (ARI 0.34) is no better than geometry alone (0.36).

Suggestive, not conclusive

Three held-out seeds, spread 0.02 to 0.06, one synthetic generator, threshold 0.3 tuned on separate seeds.

Next step

Validate on real WM811K maps and retune the threshold for their noise.

Hard variation level · held-out seeds 2001–2003 · topology uses point-cloud threshold 0.3